AahaarTwin — system architecture (static + dynamic fusion → adverse-event prediction → doctor dashboard)

1 · DATA SOURCES
CGMacros v1.0.0 — DOI 10.13026/3z8q-x658
Dexcom/Libre CGM · Fitbit (HR, activity)
meal logs (carbs/protein/fat/fiber/calories)
bio.csv static labs (age, BMI, HbA1c,
fasting glucose/insulin, lipids)

2 · PREPROCESSING — src/preprocess.py
schema normalization (8 raw variants) → dedupe 1,706 → 1,613 events
meal–CGM alignment: valid 2-h window required → 1,579 usable events, 45 patients
labels: iAUC, peak, excursion (2-h peak ≥ 180 mg/dL, rate 39.5%)

3 · FEATURES — src/features.py · 25 features
5 meal macros | 4 pre-meal dynamic (g0, 60-min glucose slope, mean HR 2 h,
activity kcal 2 h) | cyclical time-of-day | meal-type one-hot | 8 static biometrics

4 · EVALUATION HARNESS — src/splits.py, src/baselines.py
5-fold patient-level GroupKFold (no leakage: patient never in train+eval)
baselines: patient-mean · carb-only linear · population linear

5 · MODELS — src/train.py → 5 LightGBM boosters (models/)
lgbm_iauc (iAUC) · lgbm_peak (peak) · lgbm_excursion (excursion classifier)
lgbm_iauc_q05 / lgbm_iauc_q95 (quantile bounds)

6 · UNCERTAINTY — src/conformalize.py
conformalized quantile regression (α=0.1): OOF coverage 0.901 (raw: 0.605),
mean width 119.2 · isotonic calibration of excursion probs (ECE 0.122 → ~0)

7 · TWIN STATE — src/twin_state.py → twin_state.db (SQLite)
seeded by replaying 1,579 meals in order (predict-first, then observe)
per-patient profile: running residual stats, drift bound 48.58 (iAUC units)
every prediction logged with its later observed outcome

8 · MEAL RANKING — src/ranking.py → 9 · DASHBOARD — dashboard/app.py
rank candidate meals by predicted response · 7-page Streamlit app:
E2: model Spearman 0.546 vs carb-only 0.274 · Overview · Patient explorer ·
Prediction · Meal comparison · Twin history · Evaluation · Safety
all metrics loaded from models/*.json artifacts; disclaimers on every page

OOF evaluation → models/metrics.json, metrics_ci.json

predicted-vs-observed → profile updates

drift warnings → Safety page

Research prototype — not a medical device; not for diagnosis or treatment.